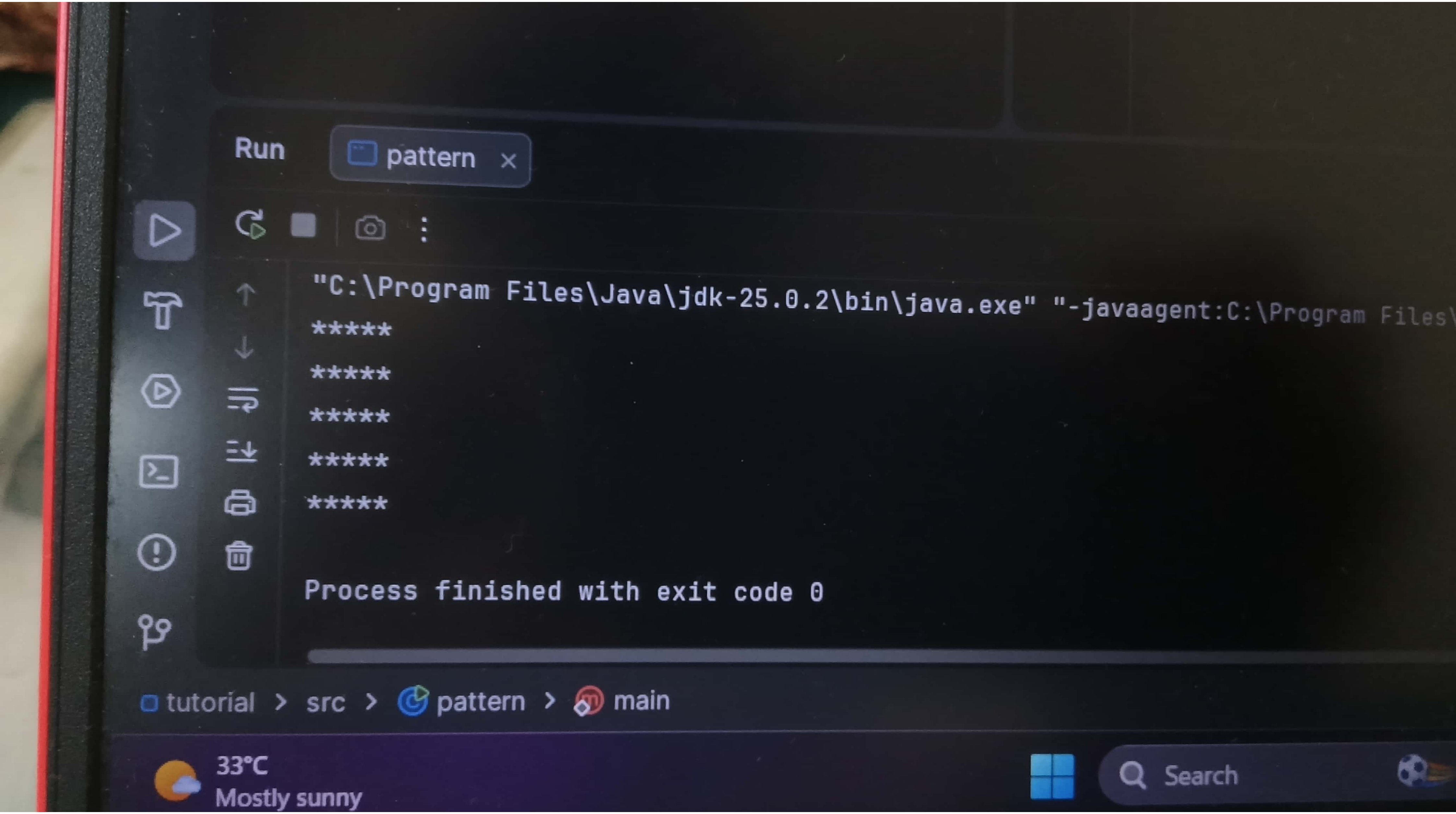
MyClass.java   hello.java   pattern.java

```java
public class pattern {
    static void main(String[] args) {

        for (int i = 0; i < 5; i++) {
            for (int j = 0; j < 5; j++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}
```


Run
pattern
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\
*****
*****
*****
*****
*****
Process finished with exit code 0
tutorial > src > pattern > main
33°C
Mostly sunny
Search

Date : __/__/____

Pattern

```
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
```

Rows = column = 5

Steps 1 - Identify the no. of rows and columns

Steps 2 - Each row and column as 5 starts

Steps 3 Convert the rows and column in a loop.

where for loop i represents rows
and for loop j represents columns

So, Outer loop = row
inner loop = column

ex -
```
for(int i=0, i<5, i++){
        for (int j=0, j<5, j++){
    System.out.print("*");
    }
    System.out.print("  ");
}
```

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Date :___/___/___

Code ÷

```
public class pattern{
    public static void main(String[] args){
    for (int i=0; i<5; i++) {
        for(int j=0; j<5; j++){
            System.out.print("*");
        }
        System.out.println(" ");
    }
  }.
}
```

ontrol

pattern

Start Free Trial

MyClass.java hello.java pattern.java

eaProjects\tutorial

```java
public class pattern {
    static void main(String[] args) {

        for (int i = 0; i < 5; i++) {
            for (int j = 0; j < 5; j++) {
                if (i==0 || i==4 || j==0 || j==4){
                    System.out.print("*");
                }
                else{
                    System.out.print(" ");
                }
            }
            System.out.println(" ");
        }

    }

}
```

```
java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=63073" -Dfile.encoding=UTF-8 -Dsun.stdo
```

```
                System.out.println("
            }

        }

    }

```

Run   pattern ×

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\I
*****
*   *
*   *
*   *
*****

Process finished with exit code 0
```

tutorial > src > pattern > main

Date : ___/___/______

Pattern

5

```
* * * * *
*       *
*       *
*       *
* * * * *
```

Logic

Step 1 No. of rows

row = column = 5

Step 2

Row 1 has 5 stars
Row 2 has 2 stars and 3 spaces
Row 3 has 2 stars and 3 spaces
Row 4 has 2 stars and 3 spaces
Row 5 has 5 stars

Step 3 Convert into loop

so as there are hollow spaces b/w the pattern, we will add if condition in the loop and print starts only in borders

like,

```
if(i==0 || i=4 || j=0 || j == 4) {
        print ("* ");   }
else{
    print ("    ").
```

Date : __/__/____

~~Code~~ ÷

~~public class pattern (String[] args) {~~

Code ÷

```
public class pattern {
    public static void main(String[] args) {
        for (int i = 0; i < 5; i++) {
            for (int j = 0; j < 5; j++) {
                ~~System.out~~
                if (i == 0 || i == 4 || j == 0 || j == 4) {
                    system.out.print("*"); }
                else {
                    system.out.print(" "); }
            }
        }
    };
}
```

Project

tutorial C:\Users\mayan\IdeaProjects\tutorial
- .idea
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  - MyClass.java
  - pattern
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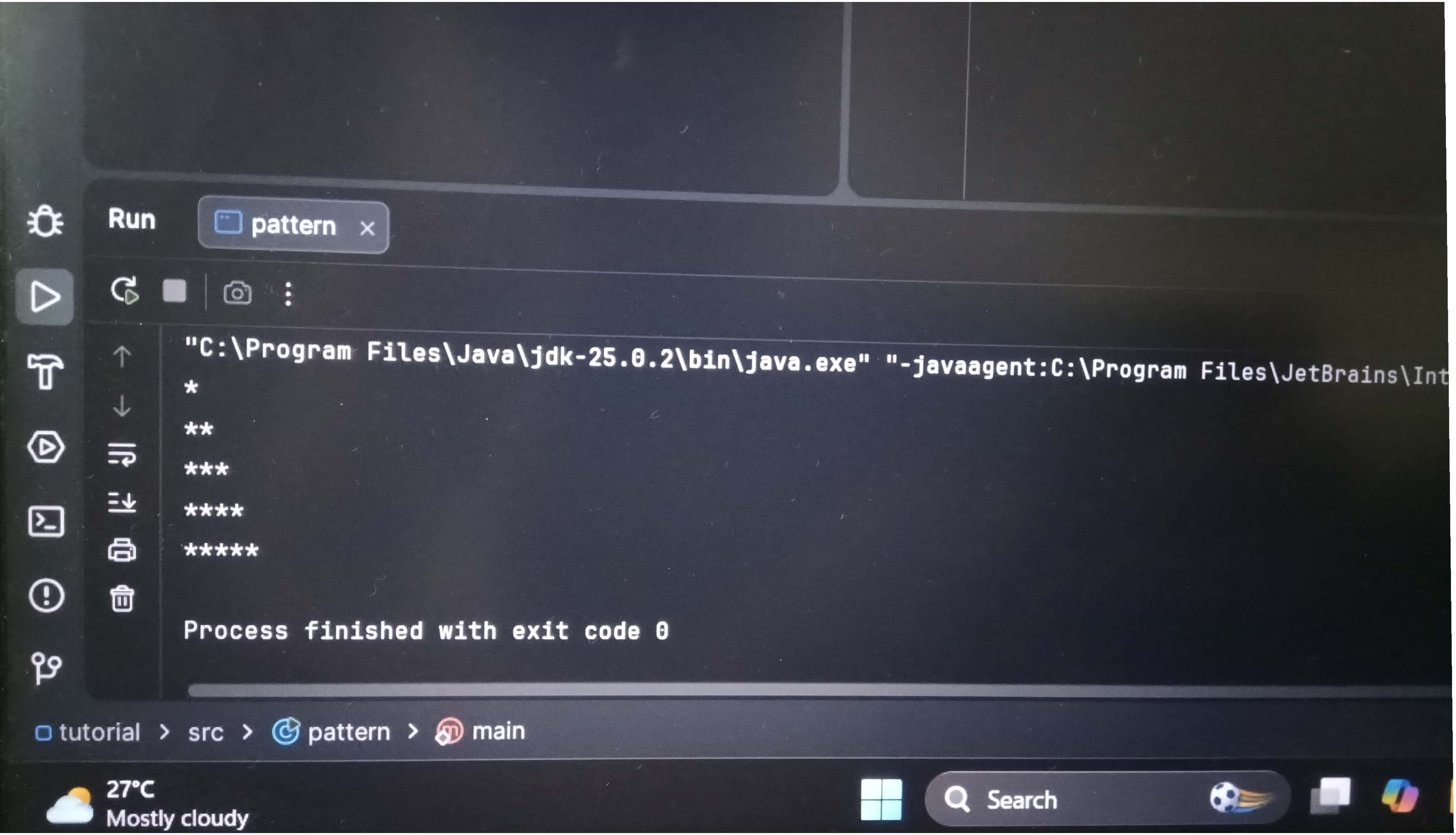
- External Libraries
- Scratches and Consoles

MyClass.java | hello.java | pattern.java

```java
public class pattern {
    static void main(String[] args) {

        for (int i = 0; i < 5; i++) {
            for (int j = 0; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}
```

Run | pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=53964" -Dfile.encoding=UTF-
*
**
***
****
*****

Process finished with exit code 0
```


Run
pattern
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\Int
*
**
***
****
*****
Process finished with exit code 0
tutorial > src > pattern > main
27°C
Mostly cloudy
Search

Date : __/__/____

Pattern 5

```
*
* *
* * *
* * * *
* * * * *
```

Logic

Step-1 No. of rows = 5
No. of columns = 5

Step2 Row number = no. of stars prints

ex- Row 1 = 1 star ('*')
Row 2 = 2 star ('* *')

Step3 Convert into loop

Outer loop will works from i = 0 to 4 but inner loop will run from j = 0 to i for each row.

Date : __/__/__

Code :-

```
public static pattern {
    public static void main(String[] args){
    for (int i=0 ; i<5 ; i++){
            for( int j=0; j<=i ; j++){
                System.out.print('*');
            }
        System.out.println(" ");
        }
    }
}
```

tutorial C:\Users\mayan\IdeaProjects\tutorial
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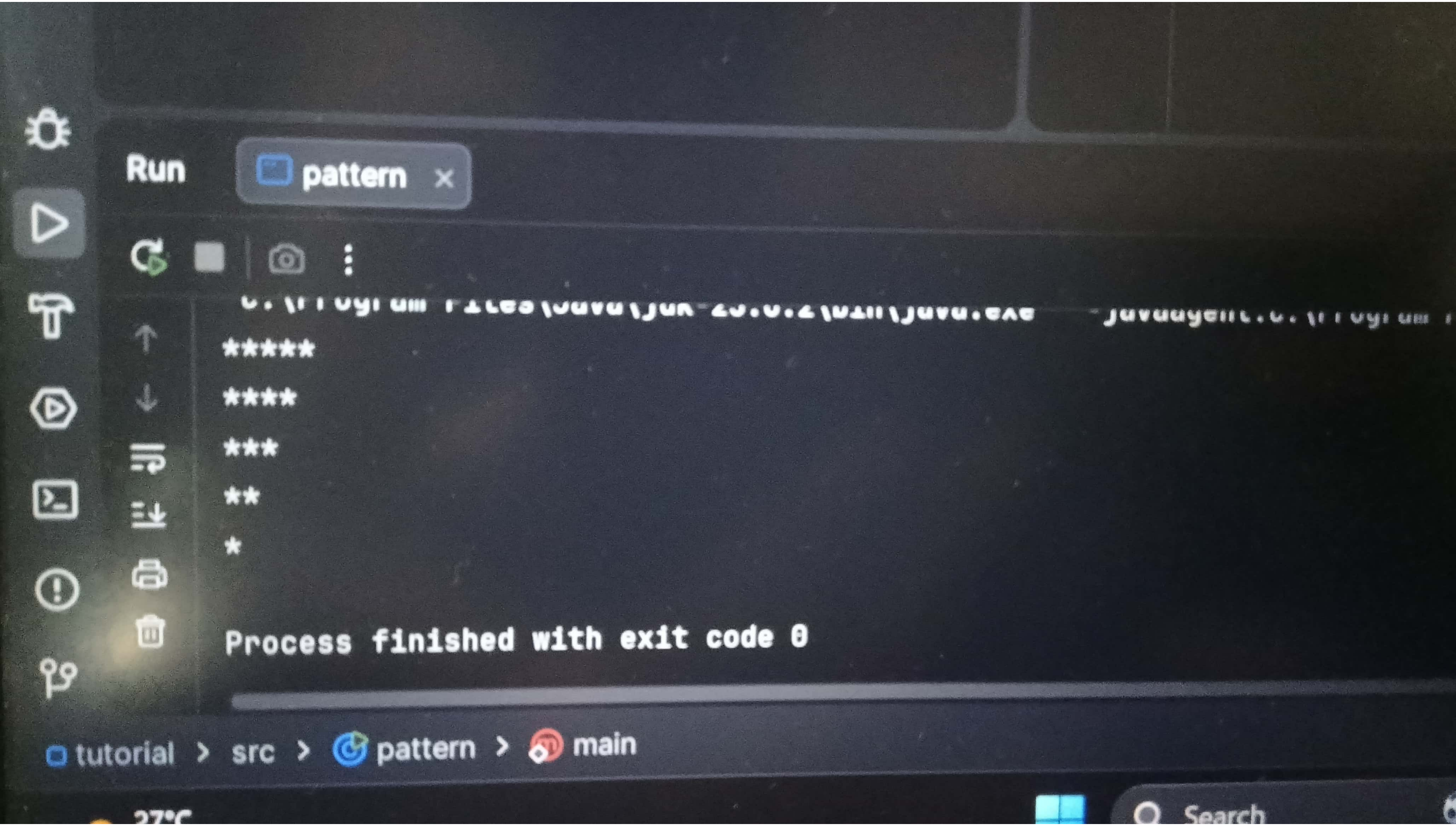
MyClass.java | hello.java | pattern.java

```java
public class pattern {
    static void main(String[] args) {

        for (int i = 5; i > 0; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}
```

Run | pattern

```
*****
****
***
**
*
```


Run
pattern
*****
****
***
**
*
Process finished with exit code 0
tutorial > src > pattern > main
Search

Date : __ / __ / ____

Pattern

* * * * *
* * * *
* * *
* *
*

logic

Step 1 Rows = 5
Columns = 5

Step 2 This time Rows are inverted as
Row 5 → stars 5
Row 4 → stars 4
Row 3 → stars 3
Row 2 → stars 2
Row 1 → stars 1

Step 3 Convert to loop

Now the outer loop will run from i=5 to 1 and inner loop will be same run from j=1 to i.
In which the inner loop prints the number of stars.
So the out loop will run in reverse or decreasing order.

Date : __/__/__

Code ÷

```
public class pattern {
    public static void main (String[] args){
        for (int i = 5; i > 0; i--) {
            for(int j = 1 ; j <= i ; j++) {
                System.out.print('*');
            }
            System.out.println(' ');
        }
    }
}
```

j = 1 4

j = 2 3

j = 3 2

j = 4 1

j = 4 0

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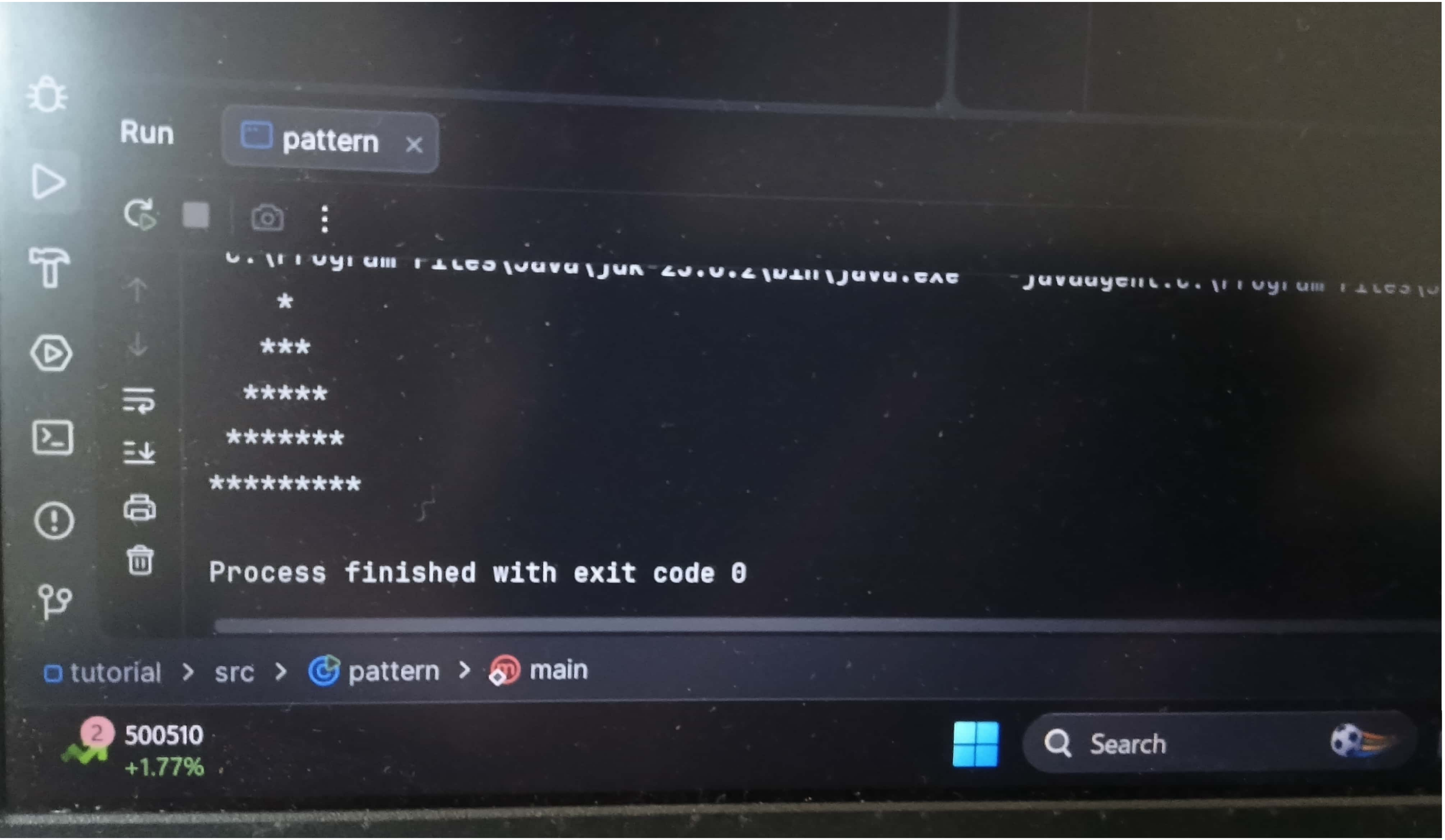
MyClass.java   hello.java   pattern.java

```java
public class pattern {
    static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5-i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2*i-1; k++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}
```


Run
pattern
    *
   ***
  *****
 ****
****
Process finished with exit code 0
tutorial > src > pattern > main
500510
+1.77%
Search

Date : __/__/____

Pattern

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
```

Logic

Step-1 Row = 5

Step-2

Row 1 = 4 spaces 1 star

Row 2 = 3 spaces 3 star

Row 3 = 2 spaces 5 star

Row 4 = 1 space 7 star

Row 5 = 0 space 9 star.

Step-3 convert to loop

There will be one outer loop which runs from i = 1 to 5 and there will be two inner loop :

loop 1 for spaces

loop 2 for stars

loop 1 runs from j = 1 to 5 − i

loop 2 runs from k = 1 to 2 * i − 1.

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Code ÷

```
public class pattern {
    public staic void main (String[] args){
        for(int i=1 ; i<=5 ;i++){
            for(int j=1 ; j<=5-i; j++){
                System.out.print(' ');
            }
            for (int k=1; k<= 2*i-1 ; k++){
                System.out.print('*');
            }
            System.out.println(' ');
        }
    }
}
```

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tutorial Version Control pattern Start Free Trial

Project

- tutorial C:\Users\mayan\IdeaProjects\tutorial
  - .idea
  - out
  - src
    - hello
    - Main.java
    - MyClass.java
    - pattern
  - .gitignore
  - tutorial.iml
- External Libraries
- Scratches and Consoles

MyClass.java hello.java pattern.java

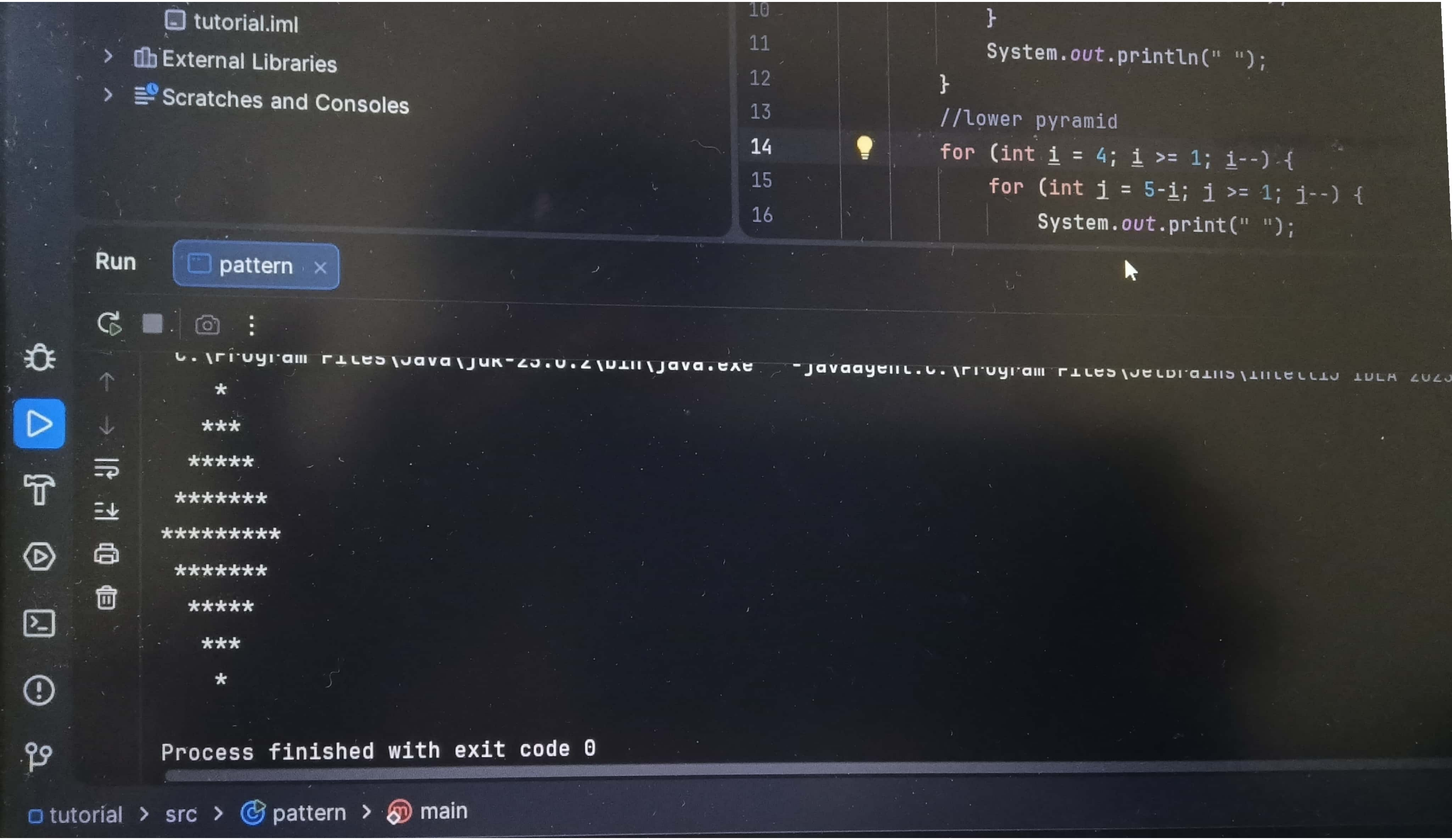
```java
public class pattern {
    static void main(String[] args) {
        // upper pyramid
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5-i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2*i-1; k++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }
        //lower pyramid
        for (int i = 4; i >= 1; i--) {
            for (int j = 5-i; j >= 1; j--) {
                System.out.print(" ");
            }
            for (int k = 2*i-1; k >= 1; k--) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}
```

Run pattern

ial > src > pattern > main 14:23 CRLF UTF-8 4 spaces


tutorial.iml
External Libraries
Scratches and Consoles
11 System.out.println(" ");
12 }
13 //lower pyramid
14 for (int i = 4; i >= 1; i--) {
15 for (int j = 5-i; j >= 1; j--) {
16 System.out.print(" ");
Run
pattern
*
***
*****
****
****
****
*****
***
*
Process finished with exit code 0
tutorial > src > pattern > main

Pattern :

```
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
 * * * * * * *
  * * * * *
   * * *
    *
```

Step - 1 Row = 9 ~~or~~
~~Column~~
or
There are two pyramid :
① one is normal pyramid
② other one is inverted pyramid

~~so~~ So, the upper pyramid have 5 Rows and lower pyramid have 4 rows.

Step. 2 : Upper Pyramid
Row 1 has 4 space 1 star
Row 2 has 3 space 3 star
Row 3 has 2 space 5 starr
Row 2 has 1 space 7 star
Row 1 has 0 space 9 star

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In lower pyramid,

we have Rows in decreasing order

Row 4 has 1 space 7 stars
Row 3 has 2 space 5 stars
Row 2 has 3 space 3 stars
Row 1 has 4 space 1 star

Step 3 Convert into loop.

for upper pyramid,

there will be one outer loop which runs from i=1 to 5 and two inner loops÷
① loop 1 runs from j=1 to 5-i
② loop 2 runs from k=1 to 2×i-1

for lower pyramid,

ther will be one other loop which runs from i= 4 to 1 and two inner loops÷
① loop 1 runs from j= 5-i to 1.
② loop 2 runs from k= 2×i-1 to 1

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Date : ___/___/___

Code :-

```
public class pattern {

    public static void main(String[] args) {
        // upper pyramid
        for (int i = 1 ; i <= 5 ; i++) {
            for (int j = 1 ; j <= 5 - i ; j++) {

                system.out.print("  ");
            }
            for (int k = 1; k <= 2 * i - 1; k++) {

                System.out.print("*");
            }
        }

        // lower pyramid
        for (int i = 4 ; i >= 1 ; i--) {
            for (int j = 5 - i ; j >= 1 ; j--) {

                System.out.print("  ");
            }
            for (int k = 2 * i - 1 ; k >= 1 ; k--) {

                System.out.print("*");
            }
        }
    }
}
```

ect

tutorial C:\Users\mayan\IdeaProjects\tutorial
.idea
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Main.java
MyClass.java
pattern
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tutorial.iml
External Libraries
Scratches and Consoles

MyClass.java hello.java pattern.java

```java
public class pattern {
    static void main(String[] args) {
        // upper pyramid
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println(" ");
        }
        //lower half
        for (int i = 4; i >= 1; i--) {
            for (int j = i; j >= 1; j--) {
                System.out.print("*");
            }
            System.out.println(" ");
        }

    }

}

```

pattern

```
Process finished with exit code 0
```

21:1 CRLF UTF-8

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
*
**
***
****
*****
****
***
**
*

Process finished with exit code 0
```

Date : __/__/____

Pattern :-

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Logic

Step 1 ~~Row~~ There are two triangle ~~Upp~~ Upper and lower, which together makes the half diamond.

Upper triangle has 5 rows
lower triangle has 4 rows

Step 2 Upper triange,
Row 1 → 1 star
Row 2 → 2 star
Row 3 → 3 star
Row 4 → 4 star
Row 5 → 5 star

lower triangle or inverted triangle as it has decreasing ~~5~~ rows.
Row ~~5~~ 4 → 4 star

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Row 3 → 3 star

Row 2 → 2 star

Row 1 → 1 star

So, Row number = Number of stars just like the triangle pattern.

Step-3 Convert to loop

there two loop →

① Outer loop

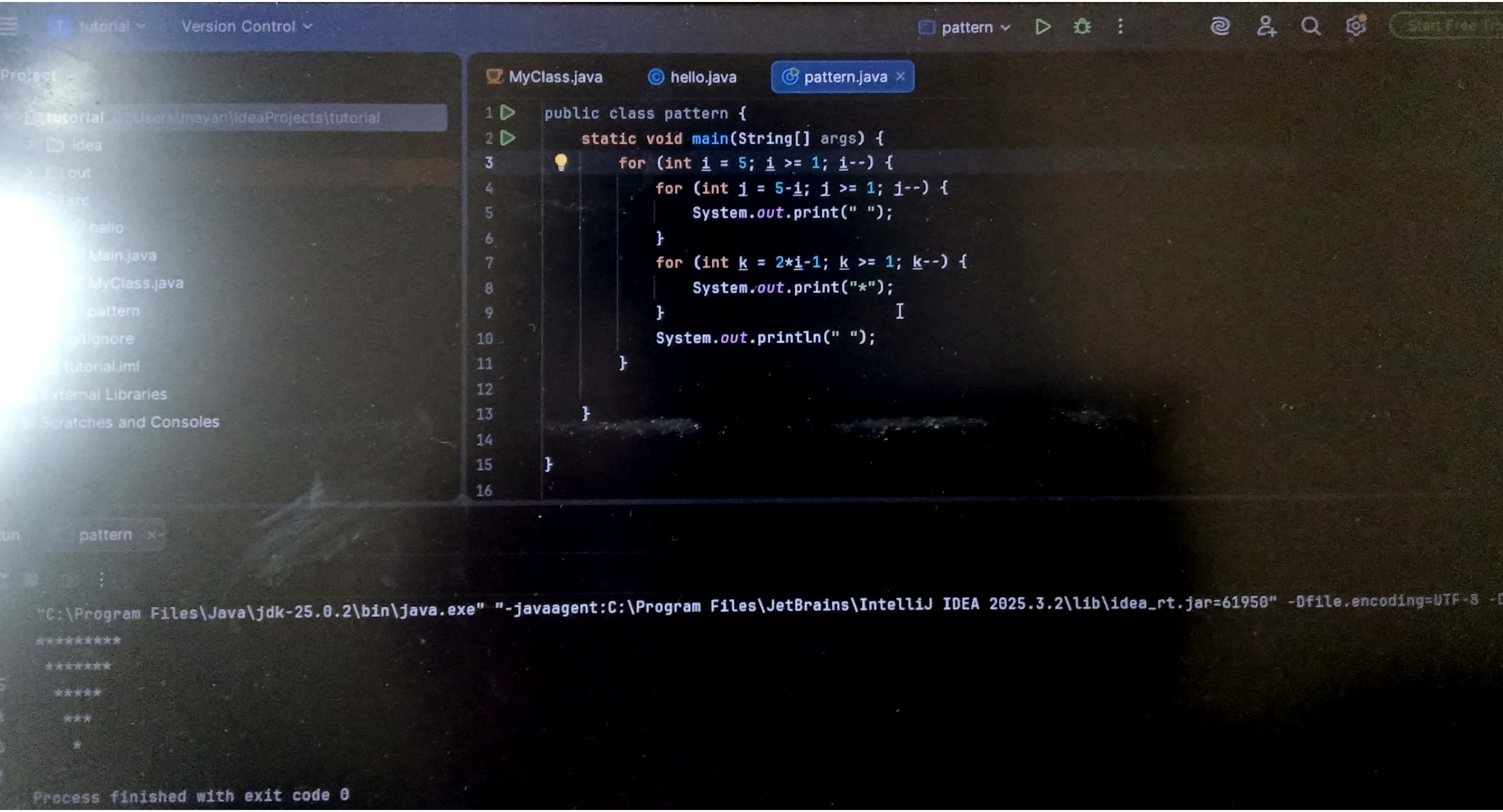
② Inner loop

for both upper triangle and lower triangle,

so, usually there are 4 loops.

Code:

```
public class pattern {
    public static void main(String[] args){
        // Upper half diamond
        for(int i=1 ; i<=5; i++){
            for(int j=1; j<=i ;j++){
                System.out.print("*");
            }
        }
    // lower half diamond
        for(int i=4; i>=1 ; i--){
        for(int j=i ; j>=1; j--){
            System.out.print("*");
        }
        }
    }
}
```


tutorial
Version Control
pattern
MyClass.java
hello.java
pattern.java
public class pattern {
static void main(String[] args) {
for (int i = 5; i >= 1; i--) {
for (int j = 5-i; j >= 1; j--) {
System.out.print(" ");
}
for (int k = 2*i-1; k >= 1; k--) {
System.out.print("*");
}
System.out.println(" ");
}
}
}
.idea
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External Libraries
Scratches and Consoles
pattern
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=61950" -Dfile.encoding=UTF-8
****
****
*****
***
*
Process finished with exit code 0

```
}
```

Run   pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\J
*********
 *******
  *****
   ***
    *

Process finished with exit code 0
```

Date : __ / __ / ____

Pattern :-

```
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

Step-1 Identify pattern,

It is a inverted pyramid or lower half diamond.

It has 5 Rows and 5 columns.

In each row the no. of space is Increasing

Step-2 The row number is in decreasing order.

Row → 5
Row → 4
Row → 3
Row → 2
Row → 1

Step 3 Convert to loop,

The will be one outer loop and two inner loop.

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Inner loop 1 will print spaces which runs from j = 5 - i to 1

~~For~~ Inner loop 2 will print stars which runs from k = 2 x i - 1 to 1.

Code :-

```
public class pattern {
    public static void main(String [] args) {
        for (int i = 5 ; i >= 1 ; i--) {
            for (int j = 5-i; j >= 1 ; j--) {
                System.out.print("  ");
            }
            for (int k = 2*i-1 ; k >= 1 ; k--) {
                System.out.print("*");
            }
            System.out.println("  ");
        }
    }
}
```

Project

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MyClass.java | hello.java | pattern.java

```java
public class pattern {
    static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5; j++) {
                System.out.print("1");
            }
            System.out.println(" ");
        }
    }

}
```

Run pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=62262" -Dfil
11111
11111
11111
11111
11111

Process finished with exit code 0
```

Run pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Progra
11111
11111
11111
11111
11111

Process finished with exit code 0
```

Date : __/__/____

Pattern :

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

logic

Step1 : Rows = 5 , colum = 5

Step2 : Outer loop for rows
and inner loop for column

Step3 : Print 1 in every column

Step4 : move in next after each row
using (println).

Code :

```
public class pattern{
    public static void main (String[] args){

    for(int i =1 ; i <=5; i++){
        for(int j = 1; j<=5;j ++ ){
            System.out.print("1");
        }
    System.out.println(" ");
    }
  }
}
```

```java
public class pattern {
    static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5; j++) {
                System.out.print(j);
            }
            System.out.println(" ");
        }
    }
}
```

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=62370" -Dfile.encoding=U
12345
12345
12345
12345
12345

Process finished with exit code 0
```

Run pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\I
12345
12345
12345
12345
12345

Process finished with exit code 0
```

tutorial > src > pattern

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Pattern

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

Logic

Step-1 Number of rows-5, column-5

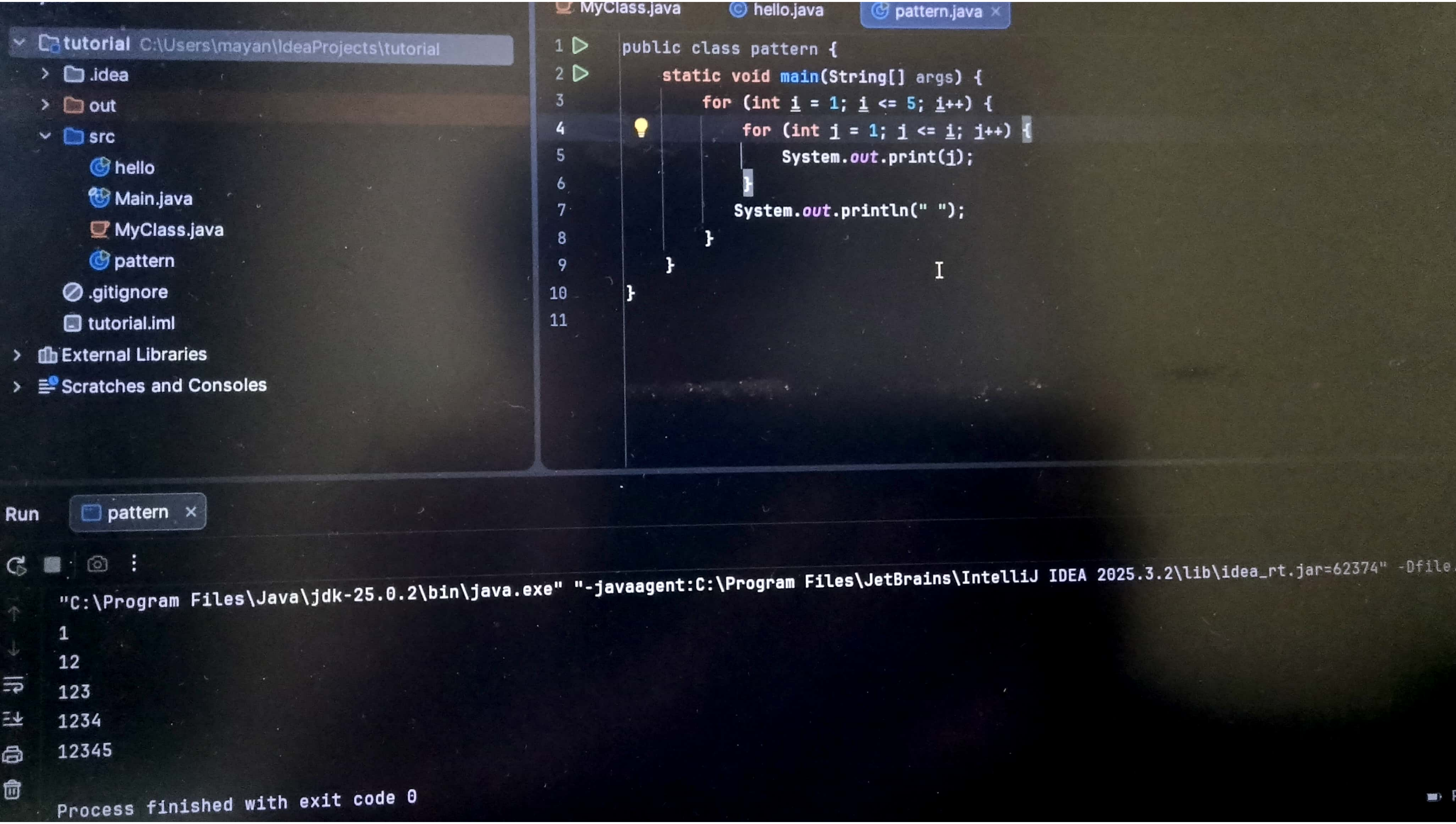
There will be two loops,
outer loop for rows
inner loop for columns.

Step 2 Print the patterns by print the value of j as it change from 1 to 5.

Step 3 Move to next line after every row.

Code :

```
public class pattern{
    public static void main(String[] args){
        for(int i=1 ; i<=5 ; i++){
            for(int j=1; j<=5 ; j++){
                System.out.print(j);
            }
            System.out.println(" ");
        }
    }
}
```


MyClass.java
hello.java
pattern.java
tutorial C:\Users\mayan\IdeaProjects\tutorial
.idea
out
src
hello
Main.java
MyClass.java
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public class pattern {
static void main(String[] args) {
for (int i = 1; i <= 5; i++) {
for (int j = 1; j <= i; j++) {
System.out.print(j);
}
System.out.println(" ");
}
}
}
Run
pattern
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=62374" -Dfile.
1
12
123
1234
12345
Process finished with exit code 0

Run   pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program
1
12
123
1234
12345

Process finished with exit code 0
```

Date : __/__/__

Pattern

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

~~Step~~
Logic ⇒

Step-1 Row = 5, colum = 5

Step-2 Outer loop for row, inner loop for column.

Step-3 The outer loop will run from i = 1 to 5 while inner will run from j = 1 to i

Step-4 print the value of j to get the pattern.

Code ÷

```
public class pattern{
    public static void main(String[] args){

    for(int i=1 ; i<=5 ; i++){

        for(int j=1; j<=i; j++){
            System.out.print(j);
        }
        System.out.println(" ");
    }
  }
}
```

ject

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  - Main.java
  - MyClass.java
  - pattern
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MyClass.java | hello.java | pattern.java

```java
public class pattern {
    static void main(String[] args) {
        for (int i = 5; i >= 1; i--) {
            for (int j = 5-i; j >= 1; j--) {
                System.out.print(" ");
            }
            for (int k = 2*i-1; k >= 1; k--) {
                System.out.print(8-i);
            }
            System.out.println(" ");
        }
    }
}
```

pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=64516" -Dfile.encodin
333333333
 4444444
  55555
   666
    7

with exit code 0
```

.gitignore
tutorial.iml
External Libraries
Scratches and Consoles

```
10            System.out.println(" ");
11        }
12    }
13 }
14
```

Run   pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 202
333333333
 4444444
  55555
   666
    7

Process finished with exit code 0
```

Date :__/__/__

Pattern :-

```
333333333
 4444444
  55555
   666
    7
```

Logic :-

Step 1 : No. of rows = 5

Step 2 There we need three loop :-
↳ Outer loop for rows.
↳ Inner loop 1 for print spaces
↳ Outer loop 2 for print number.

Step 3 : The ~~inner~~ outer loop will run from i = 5 to 1
and inner loop 1 will run from j = 5-i to 1
and inner loop 2 will run from j = 2*i-1
~~to~~ to 1.

Step 4 : print (8-i) to print the pattern.

Date : __/__/____

```
public class pattern {
    public static void main(String[] args){
    for(int i=5 ; i>=1 ; i--){
        for (int j= 5-i ; j>=1 ; j--){
            System.out.print("   ");
        }
        for(int k= 2*i-1 ; k>= 1; k--){
            System.out.print(8-i);
        }
    }
  }
}
```

Project

- tutorial C:\Users\mayan\IdeaProjects\tutorial
  - .idea
  - out
  - src
    - hello
    - Main.java
    - MyClass.java
    - pattern
  - .gitignore
  - tutorial.iml
- External Libraries
- Scratches and Consoles

MyClass.java | hello.java | pattern.java

```java
public class pattern {
    static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i+2);
            }
            System.out.println(" ");
        }
        for (int i = 4; i >= 1; i--) {
            for (int j = i; j >= 1; j--) {
                System.out.print(i+2);
            }
            System.out.println(" ");
        }
    }
}
```

Run pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=49487" -Dfile.encoding=UTF-8 -Dsu
3
44
555
6666
77777
6666
555
44
3
```

2:3 CRLF UT

Scratches and Consoles

```
            System.out.println(" ")
        }
    }
}
```

Run   pattern ×

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\In
3
44
555
6666
77777
6666
555
44
3

Process finished with exit code 0
```

Date : __/__/____

Pattern :-

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

Logic :-

Step 1 : there are two triangle upper and lower triangle.

Upper triangle has 5 rows and lower triangle has 4 rows.

Step 2 : These four loops,

2 nested loops for upper triangle and
2 nested loops for lower triangle.

In both, the outer loop is for rows and inner for column.

Step 3 : Both triangle values are printed by i+2.

Date : __/__/__

Code :-

```
public class pattern {
    public static void main (String[] args){
    for (int i=1; i<= 5; i++){
        for(int j=1; j<= i; j++){
            System.out.print(i+2);
        }
        System.out.println(" ");
    }
    for (int i=4; i>= 1; i--){
        for(int j=i; j>=1; j--){
            System.out.print(i+2);
        }
        System.out.println(" ");
    }
  }
}
```

MyClass.java hello.java pattern.java ×

```java
public class pattern {
    static void main(String[] args) {
        for (int i = 5; i >= 3; i--) {
            for (int j = 5-i; j >= 1; j--) {
                System.out.print(" ");
            }
            for (int k = 2*i-1; k >= 5; k--) {
                System.out.print(8-i);
            }
            System.out.println(" ");
        }
        for (int i = 1; i <= 2; i++) {
            for (int j = 1; j <= 2-i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2*i+1; k++) {
                System.out.print(5-i);
            }
            System.out.println(" ");
        }
    }
}
```

Run   pattern

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files
33333
 444
  5
 444
33333

Process finished with exit code 0
```

Date : __ / __ / ____

Pattern :-

```
3 3 3 3 3
  4 4 4
    5
  4 4 4
3 3 3 3 3
```

Logic :-

Step I: There are two pyramid, the uppere is inverted pyramid and lower is normal pyramid.

upper pyramid has 3 rows and lower pyramid has 2 rows.

Step 2: ~~Rows~~ for upper pyramid,

Row 1 has five 3 and 0 spaces
Row 2 has three 4 and 1 space
Row 3 has one 5 and 2 spaces

for lower pyramid
Row 1 has 1 spaces and three 4.
Row 2 has 0 spaces and five 3.

Step 3: Print the upper pyramid value by printing 8-i and print the lower pyramid values by printing 5-i.

Date : __/__/____

Code :-

```
public class pattern{
    public static void main(String[] args){
        for(int i= 5; i>=3; i--){
            for(int j= 5-i; j>=1; j--){
                System.out.print(" ");
            }
            ~~System.out.println(" ");~~
            for(int k = 2*i-1; k>=5; k--){
                System.out.print(8-i);
            }
            System.out.println(" ");
        }

        for(int i= 1; i<=2; i++){
            for(int j=1; j<=2-i; j++){
                System.out.print(" ");
            }
            for(int k=1; k<=2*i+1; k++){
                System.out.print(5-i);
            }
            System.out.println(" ");
        }
    }
}
```